

Diego Armando Urrea Méndez

Predoctoral Researcher in Statistical Hydrology

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RESEARCH PROFILE

Curious by nature and someone who values collaboration as much as the research itself, I am defending my Ph.D. at the University of Cantabria in July 2026, where five years of work on extreme precipitation, hydrological extremes, and compound flooding have taught me that going deep into problems is what leads to results that hold. I also believe that explaining complex methods clearly is part of doing good science, and it is something I genuinely enjoy. I am now looking for a postdoctoral position where I can keep learning, contribute with what I know, and be part of work that matters.

EDUCATION

Ph.D. in Coastal Engineering, Hydrobiology and Management of Aquatic Systems 2021 – Present

IHCantabria / Universidad de Cantabria · Santander, Spain

Thesis: Non-Stationary and Multivariate Techniques for Analyzing Precipitation and Flood Impacts

Director: Manuel del Jesus · Defence: July 2026

M.Sc. Integrated Water Systems Management 2019 – 2020

E.T.S. Ingenieros de Caminos, Canales y Puertos – Universidad de Cantabria · Santander, Spain

GPA: 9.1 / 10 · 62 ECTS · Award: Best Academic Record of the Cohort

Thesis: Evaluation of climate change effects on flooding in Corrales de Buelna

Directors: Manuel del Jesus & Salvador Navas

B.Eng. Civil Engineering 2010 – 2015

Escuela Colombiana de Ingeniería Julio Garavito · Bogotá, Colombia

GPA: 3.8 / 5.0 · 185 credits

RESEARCH EXPERIENCE

Predoctoral Researcher 2021 – Present

IHCantabria – Fundación Instituto de Hidráulica Ambiental de Cantabria · Santander, Spain

- Non-stationary analysis of extreme precipitation across mainland Spain using time-varying GEV distributions applied to monthly and seasonal rainfall series (HSJ 2023; Ingeniería del Agua 2025).
- Multivariate framework for spatially compounding design storms using vine copulas and Gaussian Process Regression to characterise high-dimensional joint return periods; applied to the Besaya River basin (Journal of Hydrology 2026).
- Multivariate characterisation of flood hydrographs through joint modelling of peak discharge, volume, duration and time to peak using vine copulas, with sequential analysis from bivariate to tetravariate configurations to quantify the dimensional effect.
- Synthetic Design Hydrograph generation combining tetravariate probabilistic modelling with non-parametric morphological reconstruction via Kernel Density Estimation.
- Surrogate modelling for direct flood impact return period estimation using Hybrid Downscaling Techniques, bypassing the full rainfall–runoff–inundation chain.
- Bivariate threshold framework for compound flood early-warning systems applied to the Santoña estuary (Ingeniería del Agua 2025; Coastal Engineering 2025).
- Participant in projects SATURNALIA and WaMA-WaDiT (PCI2024-153483, MICIU/AEI/UE): development of mathematical and statistical techniques for climate hazard quantification and impact assessment.

- Implemented machine-learning models (Random Forests, Linear Regression, SARIMAX) for hydro-meteorological prediction using Python.
- Applied statistical downscaling and stochastic climate variable generation for climate change scenarios.
- Prepared a technical report on climate extreme indicators as part of a European research project (ERA4CS network).

PUBLICATIONS

Published — First Author

- [J3] Urrea Méndez, D., Gómez-Rave, D.V., del Jesus, M. (2026). Multivariate design storms using vine copulas and Gaussian process regression. *Journal of Hydrology*, 135762. <https://doi.org/10.1016/j.jhydrol.2026.135762>
- [J2] Urrea Méndez, D., del Jesus, M. (2025). Modelización de precipitación extrema en España: un enfoque no estacionario integrando teleconexiones climáticas. *Ingeniería del Agua (IWA)*, 29(3). <https://doi.org/10.4995/ia.2025.23034>
- [J1] Urrea Méndez, D., del Jesus, M. (2023). Estimating extreme monthly rainfall for Spain using non-stationary techniques. *Hydrological Sciences Journal*, 68(7), 903–919. <https://doi.org/10.1080/02626667.2023.2193294>

Published — Collaborative

- [C2] Gómez-Rave, D.V., Urrea Méndez, D., del Jesus, M. (2025). Multivariate design events for compound flooding analysis in estuaries. *Coastal Engineering*, 202, 104850. <https://doi.org/10.1016/j.coastaleng.2025.104850>
- [C1] Gómez-Rave, D.V., Urrea Méndez, D., del Jesus, M. (2025). Inundación compuesta en el estuario de Santoña: umbrales bivariados y su aplicación en alertas tempranas. *Ingeniería del Agua*, 29(2), 73–89. <https://doi.org/10.4995/ia.2025.23025>

Preprints

- [P1] Urrea Méndez, D., del Jesus, M., Gómez-Rave, D.V. (2024). Return period of high-dimensional compound events. Part II: Analysis of spatially-variable precipitation. Preprint.
- [P2] del Jesus, M., Urrea Méndez, D., Gómez-Rave, D.V. (2024). Return period of high-dimensional compound events. Part I: Conceptual framework. Preprint.

CONFERENCE PRESENTATIONS

Oral Presentations (14 total)

2026

- [O14] Urrea Méndez, D., Gómez-Rave, D.V., del Jesus, M. (2026). Surrogate modelling for fluvial multivariate flood-frequency in a mountain catchment. HIC 2026 – Hydroinformatics Conference, Zaragoza, Spain.

2025

- [O13] del Jesus, M., Navas, S., Urrea Méndez, D. (2025). Comparison of fitting methods for extreme precipitation distributions: analysis of the October 2024 event in Valencia. VIII Jornadas de Ingeniería del Agua, Zaragoza, Spain.
- [O12] Gómez-Rave, D.V., Urrea Méndez, D., del Jesus, M. (2025). From statistical dependence to physical simulation: an integrated strategy for compound flooding. VIII Jornadas de Ingeniería del Agua, Zaragoza, Spain.
- [O11] del Jesus, M., Navas, S., Urrea Méndez, D. (2025). An analysis of return levels of Valencia's 2024 extreme rainfall. EGU General Assembly, Vienna, Austria.
- [O10] Urrea Méndez, D., Gómez-Rave, D.V., del Jesus, M. (2025). Advancing flood impact estimation: comparing multivariate continuous hydrologic-hydraulic models with traditional approaches. EGU General Assembly, Vienna, Austria.
- [O9] Gómez-Rave, D.V., Urrea Méndez, D., Scolobig, A., del Jesus, M. (2025). Understanding and preparing for compound flooding. EGU General Assembly, Vienna, Austria.

2024

- [O8] Gómez-Rave, D.V., Urrea Méndez, D., del Jesus, M. (2024). Advanced modelling of compound flooding in estuaries. 3rd International Conference on Natural Hazards and Risks in a Changing World, Amsterdam, Netherlands.
- [O7] Gómez-Rave, D.V., Urrea Méndez, D., del Jesus, M. (2024). Exploring compound flood hazards: bridging statistics and hydrodynamics in estuaries. 8th IAHR Europe Congress, Lisboa, Portugal.
- [O6] Urrea Méndez, D., Gómez-Rave, D.V., del Jesus, M. (2024). Multivariate models for flood prediction: a glimpse into the future. 8th IAHR Europe Congress, Lisboa, Portugal.

[05] Gómez-Rave, D.V., Urrea Méndez, D., del Jesus, M. (2024). Understanding compound flooding hazard in estuaries: insights and implications. EGU General Assembly, Vienna, Austria.

2023

[04] Urrea Méndez, D., del Jesus, M. (2023). Estimation of extreme monthly precipitation for Spain using non-stationary techniques. VII Jornadas de Ingeniería del Agua, Cartagena, Spain.

[03] Gómez-Rave, D.V., Urrea Méndez, D., del Jesus, M. (2023). Multivariate analysis of compound flooding in estuaries: application to Santoña (Cantabria). VII Jornadas de Ingeniería del Agua, Cartagena, Spain.

[02] Gómez-Rave, D.V., Urrea Méndez, D., del Jesus, M. (2023). High-dimensional multivariate analysis of compound flooding in estuaries: a case study in Santoña. IAHR Young Professionals Congress, Online.

2021

[01] Urrea Méndez, D., del Jesus, M. (2021). Non-stationary extreme rainfall analysis for Spain. ICSH-STAHY Workshop, València, Spain.

Poster Presentations (4 total)

[Po4] Urrea Méndez, D., Gómez-Rave, D.V., del Jesus, M. (2024). Exploring multivariate return periods: enhancing accuracy in hydrological analysis for flood prediction. EGU General Assembly, Vienna, Austria.

[Po3] Gómez-Rave, D.V., Urrea Méndez, D., del Jesus, M. (2023). Multivariate probability analysis of compound flooding dynamics. EGU General Assembly, Vienna, Austria.

[Po2] Gómez-Rave, D.V., Urrea Méndez, D., del Jesus, M. (2022). Multivariate analysis framework for estuarine flooding risk. 12th International Workshop on Statistical Hydrology (STAHY), Sardinia, Italy.

[Po1] Urrea Méndez, D., del Jesus, M., Gómez-Rave, D.V. (2022). Modeling multivariate extreme rainfall events using different families of copulas. STAHY 2022, Sardinia, Italy.

SCIENTIFIC SERVICE

Ad-hoc ReviewerIngeniería del Agua (IWA Publishing, UPV)

AWARDS & RECOGNITION

2020	Best Academic Record · M.Sc. Integrated Water Systems Management, Universidad de Cantabria
2010	Academic Excellence Scholarship (x2, semesters 1 & 2) · Universidad Católica de Colombia, Bogotá

PRIOR ENGINEERING EXPERIENCE

Category I Consulting Engineer — Supervisory

CONSTRUCTEV – Constructora Teatino Vargas · Colombia

– Technical, financial and environmental coordination of infrastructure projects: scheduling (Gantt, CPM/PERT), budgeting, hydraulic and geotechnical design supervision.

Jan 2019 – Jul 2019

Category I Consulting Engineer

CONSTRUCTEV – Constructora Teatino Vargas · Colombia

– Hydraulic, structural and geotechnical design for Master Sewerage Plan: topographic surveys, network design, WWTP, pumping stations, flood-risk reports.

Jul 2017 – Dec 2018

Design Engineer

Germán Monsalve Sáenz S.A.S · Colombia

– Hydrological and hydraulic modelling (HEC-RAS, HEC-HMS, IDF curves), road drainage design (box culverts, channels), flood-risk studies, BIM (AutoCAD Civil 3D).

Dec 2014 – Jul 2017

TECHNICAL SKILLS

Programming	Python · R · Julia · Visual Basic · HTML · Markdown · Quarto · LaTeX · Git
Statistical modelling	Extreme value analysis · Non-stationary frequency analysis · Multivariate modelling · Bayesian modelling · Machine learning
Computational text analysis	BERTopic · NLP · LLMs · Text mining
AI tools	Claude · ChatGPT · Gemini · Claude Code · Codex · MCP integration
Hydraulic modelling	HEC-RAS 2D unsteady flow · HEC-HMS · IBER
GIS / CAD	ArcGIS · QGIS · AutoCAD · AutoCAD Civil 3D (BIM)
Climate data	CMIP6 · ERA5 · Copernicus CDS · NetCDF / xarray · stochastic generation

LANGUAGES

Spanish	Native proficiency
English	B2 / C1 (CEFR) — Listening C1 · Reading C1 · Spoken B2 · Written B2

SELECTED TRAINING

2026	Claude Code Course (8 h)	<i>Platzi</i>
2026	Gemini Course (14 h)	<i>Platzi</i>
2024	SMCE – 3rd Training School on the Statistical Modelling of Compound Events (60 h · 10 CFU)	<i>Lake Como School of Advanced Studies, Italy</i>
2024	Advanced Course in Dam Break Modelling (80 h)	<i>Universidades de Valencia y de La Coruña</i>
2024	Grant Proposal Writing · Key Skills for Early Researchers · Academic Interview Preparation	<i>Charlesworth / UC</i>
2023	Data manipulation with Pandas & NumPy (11 h) · Data visualisation with Matplotlib & Seaborn (10 h)	<i>Platzi</i>
2022	Artificial Intelligence for Earth Monitoring (18 h)	<i>EUMETSAT</i>
2021	Programming for Geospatial Hydrological Applications (20 h)	<i>IHE Delft / UNESCO</i>
2021	Introduction to Data Science in Python (30 h)	<i>University of Michigan / Coursera</i>
2021	Hydroinformatics for Water Resources Sustainability (12 h)	<i>ECIJG / IHE Delft / UNESCO</i>
2021	Systematic Literature Review · Qualitative Methods · Open Science & Open Data	<i>Universidad de Cantabria</i>
2020	Introduction to Programming in Python I (20 h)	<i>Coursera / PUC Chile</i>